

Formal Report on Practical training CON3023X

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TERMS OF REFERENCE

This report is submitted in fulfilment of the practical training requirements for CON3023X as part of the BSc Construction Studies degree at the University of Cape Town. The practical training component requires students to obtain supervised experience within the built environment and is intended to supplement theoretical learning with practical exposure to professional construction environments.

The purpose of this report is to provide a comprehensive and reflective account of the practical training undertaken at H&I Construction on the Claremont IRT Phase 2A project. The report documents the nature and duration of the work undertaken, the technical and managerial activities observed and performed, and the skills and knowledge developed during the placement.

The report is based on the following requirements:

1. To obtain supervised employment within the built environment.
2. To produce a comprehensive reflective report documenting the practical training experience.
3. To evaluate the lessons and skills gained during practical training, with reference to the BSc Construction Studies degree outcomes.
4. To reflect on technical and management issues encountered during the training period.
5. To provide a detailed daily work diary recording the dates, hours and activities undertaken.
6. To provide employer confirmation of the practical training.
7. To include appropriate photographic documentation of the practical experience.

The report is intended to provide an accurate account of the student's practical training experience while demonstrating the integration of academic knowledge with professional construction practice.

SUMMARY

This report provides a comprehensive reflection on the practical training undertaken with H&I Construction on the Claremont IRT Phase 2A project. The placement provided exposure to a large-scale construction environment and allowed the student to develop a deeper understanding of construction management, Quantity Surveying, cost control, subcontractor management, site operations and professional communication.

The placement represented an important progression from the student's previous practical experiences. At Feytcon, the student was primarily exposed to hands-on construction activities including paving, brickwork and earthworks. Subsequent experience at H&I provided exposure to the administrative and contractual side of construction, including, procurement, Purchase Orders, subcontractor quotations and project documentation. The Claremont placement brought these experiences together by allowing the student to observe and participate in the interaction between physical construction and the financial and administrative systems used to manage the project.

A significant area of learning was the role of the Quantity Surveyor in controlling project expenditure. The student was exposed to concrete delivery reconciliations, measurements of completed concrete and formwork, subcontractor claims, payment forecasting and the verification of work completed before payment. This demonstrated that Quantity Surveying extends beyond working with quantities and spreadsheets, requiring physical verification of work and careful financial control.

The placement also contributed substantially to the student's professional development. Increased responsibility improved confidence, while communication with subcontractors developed assertiveness and professional communication skills. Attendance at toolbox talks demonstrated the importance of open communication, safety culture and teamwork within construction management.

The experience also highlighted areas requiring further development. A particularly challenging task involved compiling and restructuring concrete quantities from drawings. The difficulty resulted partly from an inadequate initial approach to organising information, teaching the student the importance of planning, systematic data collection and breaking complex tasks into manageable sections.

Overall, the placement strengthened the student's respect for construction and its management and increased his desire to gain longer-term exposure to the industry. While maintaining an interest in Quantity Surveying, the experience particularly strengthened his desire to become more involved in construction management and live site operations.

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1. INTRODUCTION

1.1 Subject of and Motivation for Report

This report documents the practical training undertaken by Sage Agnew, a final-year BSc Construction Studies student at the University of Cape Town, during employment with H&I Construction on the Claremont IRT Phase 2A project in 2026.

The motivation for this report stems from the requirement for Construction Studies students to complete supervised practical training within the built environment. The practical training component provides an opportunity to supplement theoretical learning through exposure to professional construction environments.

The placement was particularly valuable because it provided exposure to a large-scale live construction project and allowed the student to interact with construction professionals while undertaking tasks associated with both construction management and Quantity Surveying.

1.2 Background to Report

UCT's Construction Studies programme requires students to complete practical training in accordance with CON1007X, CON2013X and CON3023X. The practical training requirement is intended to provide students with experience within the built environment and allow theoretical knowledge gained during the degree to be applied in professional circumstances. The student's previous practical experience included hands-on construction work at Feytcon and office-based exposure at H&I. The Claremont IRT placement provided a further progression by placing the student within a large-scale live construction environment. This placement therefore provided an opportunity to understand the relationship between physical construction activities, site management, subcontractors, Quantity Surveying, cost control and project administration.

1.3 Objectives of Report

The objectives of this report are to:

- Document the activities and tasks undertaken during the practical training period.
- Provide a reflective analysis of the skills and knowledge acquired.
- Link practical experiences to the BSc Construction Studies degree outcomes.
- Evaluate the relationship between theoretical knowledge and professional practice.
- Reflect on technical and managerial issues encountered during the placement.
- Document the student's professional development and future career interests.
- Provide a detailed daily work diary of the practical training period.

1.4 Limitations and Scope of Report

The report covers the student's practical training experience at H&I Construction's Claremont IRT Phase 2A project.

The scope includes exposure to:

- Site operations and construction activities.
- Quantity Surveying.
- Concrete measurement and reconciliation.
- Formwork measurements.
- Subcontractor claims and payment verification.
- Payment forecasting.
- Drawing interpretation and quantity measurement.
- Subcontractor communication.
- Toolbox talks and site safety.
- Quality-control testing, including DCP and concrete slump testing.
- Site inspections and observations.

A significant limitation of the placement was its relatively short duration. Although the student gained exposure to numerous activities, a short placement cannot provide the same depth of experience as remaining on a project for several months or years. The student was therefore exposed to selected stages and activities rather than the complete project lifecycle. This limitation has influenced the student's desire to obtain longer-term industry exposure after completing his studies.

1.5 Plan of Development

The report begins with an introduction outlining the motivation, background, objectives, scope, and research approach. The second section presents a reflective essay focusing on the student's development from previous practical experiences through to the Claremont IRT placement. This essay discusses initial impressions of the site, the development of Quantity Surveying knowledge, cost control, responsibility, communication, safety, the application of academic theory, challenges encountered, and professional development. The report subsequently provides a detailed daily work diary, conclusion, reference list, AI acknowledgement, employer confirmation, and photographic documentation.

1.6 Research Approach

The report is primarily based on the student's practical experience, daily logbook, direct observations and discussions with construction professionals during the placement. The student also drew upon construction drawings, measurement information, spreadsheets, site observations and academic knowledge gained throughout the BSc Construction Studies programme.

The reflective component is based on the student's own experiences and personal responses to the practical training period. This approach is consistent with the purpose of the reflective essay, which requires students to evaluate lessons and skills learned during practical training and relate them to their academic development.

2. REFLECTIVE ESSAY

2.1 Company and Project Background

H&I Construction is the appointed main contractor for the Claremont IRT Phase 2A project, where I completed my practical training. My placement also had personal significance as H&I has supported my studies through a sponsorship arrangement since 2025. This gave me the opportunity to gain practical experience within a company that had already contributed to my academic development.

The Claremont IRT project forms part of Phase 2A of the City of Cape Town's Integrated Rapid Transit (IRT) programme. The project involves the construction of a new MyCiTi bus station and associated infrastructure intended to improve connections between several forms of public transport, including MyCiTi, GABS, the UCT Shuttle, passenger rail and minibus taxis.

The project was procured under Tender No. 120Q/2024/25, with the City of Cape Town's Urban Mobility Directorate acting as the client, FENG JV providing professional services and H&I Construction appointed as the main contractor under a CIDB Grade 9GB requirement.

The facility primarily consists of reinforced-concrete construction, including piled foundations, pile caps, columns, walls and suspended slabs. A notable feature of the design is that the substructure has been designed to accommodate a potential future 10- or 12-storey development above the facility, contributing to the extensive piling requirements across the site. The site is divided into nine construction areas, with a total of 583 piles having been completed by the time of my placement.

I worked within the project team under the guidance of Site Agent Kevin Ward and Site Quantity Surveyor Bongeka Ndyumbu. My role provided exposure to both the construction and Quantity Surveying aspects of the project. I assisted with activities including concrete measurements and reconciliations, drawing measurements, subcontractor-related work, payment forecasting and cost-control processes.

The scale of the Claremont IRT project provided me with an opportunity to apply knowledge gained throughout my Construction Studies degree within a live construction environment. It also allowed me to see how the physical work taking place on site connects with the financial, administrative and management processes required to successfully deliver a major construction project.

2.2 Reflective Essay

My practical training at H&I Construction on the Claremont IRT Phase 2A project was a significant step in my development as a Construction Studies student. Having already experienced construction through hands-on work at Feytcon and later through office-based work at H&I, I entered this placement with a much more broader understanding of what construction involved. However, being placed on a large, active construction site allowed me to experience the industry in a way that my previous placements had not. I was no longer only working with the physical construction process or observing the administrative side of a project, I was here able to see how the two connected and how the decisions made by construction professionals directly influenced what was happening on site.

My first genuine reaction when arriving at the Claremont IRT site was that this place meant serious business. I was surrounded by a huge towering crane, large excavations for piling, concrete mixers, digger loaders, compactors and other heavy construction equipment. The scale and complexity were unlike anything I had previously experienced full-time. Instead of making me nervous, however, it excited me. I always wanted to walk around and see what was happening on site because I was finally seeing how a large construction project operates around me. The fact that the site was relatively close to my residence also made travelling to work easy, but it was really the people on the project who made me feel comfortable with the new environment I was in.

I quickly started introducing myself to the people on site and then to my surprise, telling them about myself was an unexpectedly positive experience. They showed genuine interest in getting to know me, which made me feel welcomed. I also discovered that I was not the only UCT student on the project and that another student who was in my class was also here. This immediately gave me a greater sense of comfort because I had someone of a similar position within the office environment. This may appear to be a small part of my experience, but it had an important effect on my confidence. Once I felt that I belonged, I became more comfortable asking questions, taking on more tasks and communicating with the people around me. I carried one of the most important lessons from my previous practical experiences directly into this placement, which was that I did not have to pretend that I knew everything. Asking questions allowed me to understand what I was doing and, more importantly, why I was doing it.

Looking back across my three practical experiences, I can now see a clear development in my understanding of construction. At Feytcon, my experience was predominantly physical. I learned that construction is not something that can simply be observed from a distance. Working with materials, tools and labour gave me an appreciation for the physical effort required to produce a finished structure. At H&I's office, my perspective changed again. I was exposed to documentation, procurement, Purchase Orders, subcontractor quotations, contract administration and contractor Quantity Surveying measurements. I began to understand that a large part of construction takes place away from the physical site. Claremont brought these experiences together. This time, the information I was working with in spreadsheets and drawings represented physical work that was happening directly in front of me. I could see how a quantity recorded on a spreadsheet related to concrete that had actually been poured, how subcontractor claims related to completed work and how site

activities affected financial decisions. This changed my perception of the contracting Site Quantity Surveyor considerably.

Before this experience, I understood Quantity Surveying mainly through the academic and office-based aspects of measurement, estimating and contract administration. Working alongside the Site QS and Senior QS showed me the level of responsibility involved in controlling project expenditure. One statement from the Site QS that remained with me was that I should have the mindset that I should "not spend any money at all". I understood this as meaning that every payment must be carefully considered and verified. Payments for concrete, formwork and subcontractor work could not simply be accepted because they appeared on a claim. The work had to physically exist and the quantities had to be checked. This was particularly interesting to me because of the consequences of inaccurate information would lead to possible budget overruns. Subcontractors may claim for work that has not yet been completed, and if this is not identified, the contractor may pay for work that has not actually been done. The QS therefore has to maintain accurate records of what has been completed. Concrete could be tracked by recording what had been poured and checking it against drawings and reconciliation spreadsheets. For example, where a number of columns were required, the total concrete quantity could be calculated and compared against the actual concrete placed on site. This allowed the QS information to reflect what was physically happening on the project.

Being involved in this process changed the way I viewed my own work. At first, I was genuinely surprised that the site QS trusted me enough to use the information I produced. I doubted myself and became extremely careful because I knew that the information I provided needed to be correct. Over time, I realised that my measurements and observations were actually being used in the work produced by the QS. This was one of the most rewarding feelings I experienced during the placement. Knowing that my work was contributing to actual project decisions made me feel useful and gave me greater pride in what I produced.

One particular experience demonstrated this to me. I was measuring floor finishes from drawings and noticed that the tile specified on the plan did not correspond with the physical tile sample. The difference caught my attention, so I raised it with the Site Agent. He thanked me for identifying the discrepancy early and said, "Well done Boet." Although this was a small interaction, it had a large impact on my confidence. I had not simply completed a task that had been given to me, I had noticed something independently and brought it to the attention of someone who could act on it, the site Agent Kevin Ward. This showed me that attention to detail has real value in construction.

Also to relate the lesson about false claims which subcontractors often make for not completed yet, when I was discussing this with the site QS I could easily remember this being taught in class, specifically Cost Engineering or CON3043W. our lecturer Melvern often stated how this was a common issue.

The placement also taught me that Quantity Surveying requires looking forward as well as recording what has already happened. I was exposed to payment forecasting and learned how expected work needs to be considered when payments are being prepared. Subcontractors are paid according to particular payment periods, meaning that work expected to take place between the current point and the payment date must also be considered. I found this

particularly interesting because it showed me the planning and foresight required from a QS. The role is not simply about measuring what has happened it also requires anticipating what is likely to happen and ensuring that the financial side of the project is prepared accordingly.

My experience with subcontractors also changed the way I communicate professionally. Initially, I thought that sending an email requesting information would be enough. I soon realised that subcontractors have their own businesses, workloads and priorities. When I needed pricing or information from subcontractors, I sometimes had to send follow-up emails and make phone calls to ensure that my request was dealt with.

At first, I was nervous about making these calls. I worried that I might ask for the wrong information, communicate incorrectly or represent the company inappropriately. To manage this, I asked the Site Agent what information I needed and how I should approach the request. Once I began making the calls, I became more confident and assertive. With that being said I also learned that professional communication sometimes requires persistence. If information or a service is needed for another activity to proceed, simply making the request once may not be enough. This experience helped me understand that construction relies heavily on communication between different parties and that delays in communication can eventually contribute to delays in the work itself. What happened essentially was that I was tasked to contact two companies to price a Bill of quantities(BOQ) for paintwork and waterproofing so that we can compare costs and get the work done, the issue was that even after I phoned and emailed them both companies took their sweet time to get back to me with the priced BOQ. This is where the Site Agent taught me that persistence and assertiveness is sometimes very important in getting work done quickly.

Another important lesson came from attending toolbox talks. Kevin, the Site Agent, consistently insisted that I attend these meetings, and I began to appreciate why. The meetings were not simply a situation where management gave instructions to workers. They also provided an opportunity for workers to raise concerns, discuss issues and contribute their own experiences. This showed me how management's attitude towards safety and communication can influence the overall spirit of a project. And what I mean here is that management during these weekly talks created a space for everyone to openly talk freely without a feeling of guilt or pressure where they could easily voice their concerns and this gave me a sense of huge comfort, I remember thinking "this is so wholesome and this is definitely a work space I want to be a part of."

I had previously encountered the importance of communication, teamwork and professionalism through university subjects such as Construction Management and Construction Professionalism. However, being physically present in these meetings allowed me to experience their effect rather than simply understand the concept. I saw how a construction project can become a common goal where people are able to communicate openly and seek assistance when problems arise. This strengthened my respect for construction management because I began to understand that managing a project is also about managing the relationships between the people responsible for delivering it.

At the same time, some of the technical activities I encountered on site demonstrated that university theory can prepare a student effectively for practical work. The DCP tests short for The Dynamic Cone Penetrometer which is a field method used in construction to evaluate in-

situ soil strength and compaction quality. Another test we regularly did whenever a concrete delivery took place was a concrete slump test. I had already encountered these tests in subjects such as Construction Technology 1 and Building Science 1, so when I saw them being conducted on site, the information was very familiar. I understood what was being tested and why the tests were necessary. This experience reinforced the value of the academic foundation provided by my degree. The practical environment did not replace the theory but instead, it gave the theory a physical context.

Not every experience during the placement was positive or easy. Towards the end of my training, I was asked to restructure and compile concrete quantities for concrete structures and walls. The task required me to work from drawings, identify structures, obtain dimensions such as heights, widths and lengths, from these measurements I would then need to apply the appropriate measurement formulas and distinguish between work that had already been completed and work that was still to be constructed.

I initially approached the task incorrectly. I did not establish a proper system for collecting the information and measurements from the drawings and did not follow the instructions as effectively as I should have. As the task progressed, I began to feel overwhelmed because every measurement made me question whether I had done it correctly. I eventually realised that the difficulty was not simply the amount of work, but the way I had approached it. I needed to organise the information systematically, identify all the structures and ensure that I had every dimension and formula required before beginning the calculations.

This experience taught me that good technical ability is not enough on its own. The way a task is planned and organised can determine how efficiently it is completed. It also reminded me that confidence does not mean knowing everything. Bongeka, the Site QS, had previously told me that I could not learn everything within such a short period and that time and exposure were necessary to develop confidence. My experience with the concrete quantities helped me understand what she meant. I could ask questions and receive assistance, but I also needed to develop my own systems for managing information and approaching complex tasks.

Additionally, when I mentioned that I needed to use the measurements I took for heights, widths and lengths of certain structures, when I needed to produce theoretical quantities for concrete walls and other concrete structures I could directly refer my Standard System provided to me from the course Measurement, Design and Appraisal II or CON3031W II, I was also told that using this Standard System aligns directly with her measurement calculations because Bongeka showed me the BOQ and how these elements are to be measured.

Overall, I experienced significant personal and professional growth during this placement. My confidence, communication, assertiveness, attention to detail, willingness to ask questions, ability to take responsibility and ability to work independently all improved. More importantly, I developed a greater respect for construction and the management required to deliver a project successfully.

I now see these qualities as interconnected. Confidence makes it easier to ask questions, asking questions improves understanding, better understanding improves the quality of my work, and producing accurate work creates trust. That trust then gives me the confidence to take on greater responsibility. This cycle has changed the way I think about professional

development. I no longer see learning as simply acquiring individual technical skills. I see it as developing a mindset that allows those skills to grow through experience.

The experience has also influenced what I want from my future career. I want to spend longer periods working within a construction environment so that I can become accustomed to the rhythm of the industry and experience how projects develop over longer periods. A short placement can introduce me to many different activities, but longer exposure would allow me to understand how responsibilities develop, how problems evolve and how decisions made at one stage of a project affect later stages.

I remain interested in Quantity Surveying and would like to continue exploring that aspect of construction, particularly the financial and contractual responsibilities I experienced at Claremont. However, I have also developed a strong desire to be more directly involved in the construction industry itself, particularly through construction management and continued site exposure.

When I think about my three practical experiences, I see more than three separate placements. I see a progression from the physical construction work at Feytcon, through the administrative and contractual environment at H&I, and finally to the combination of these experiences on the Claremont IRT project. Each experience taught me something that prepared me for the next one.

If I could give advice to my first-year self, I would say not to be afraid to get my hands dirty, not to be afraid to ask questions, to pay attention even when I am not directly involved and to take every task seriously. I would tell myself that I do not need to know everything immediately, but I do need to remain willing to learn. I would also encourage myself to build relationships with the people around me and to take responsibility for the quality of my work. The Claremont IRT placement has therefore left me with more than technical knowledge. It has made me more confident, more curious and more eager to gain further experience. Most importantly, it has made me hungry for greater involvement in the construction industry. I now understand that the more time I spend within these environments, the more opportunities I will have to learn, contribute and develop into the construction professional I want to become.

3. CONCLUSION

My practical training at H&I Construction transformed my understanding of how construction projects are delivered. Moving from previous labour-based site experience into an environment involving project management, quantity surveying, procurement and contract administration showed me that successful construction depends on much more than physical work. Working with BOQs, Purchase Orders, workload planners and subcontractor quotations demonstrated the importance of accurate information and financial control, while site inspections and Practical Completion meetings showed me how contractual requirements influence the work ultimately delivered.

The placement also strengthened the relationship between my university studies and industry practice. Concepts from Construction Management, Measurement and Design Appraisal, Business Law and Cost Engineering became more meaningful when applied to real projects. Most importantly, I developed greater confidence in my ability to learn. I entered H&I afraid that asking questions would expose my lack of knowledge, however, left understanding that asking questions, accepting feedback and learning from mistakes are essential parts of professional development.

The experience has therefore not only expanded my technical understanding but also changed how I approach learning and responsibility. I leave the placement with a broader view of construction, a clearer understanding of the roles within the industry and an overall stronger foundation for my future career.

Reference List

- H&I Construction. (2025-2026). Company Profile and Corporate Information. H&I Construction.
- University of Cape Town. (2026). CON3023X Practical Training Guidelines. Department of Construction Economics and Management.

AI Acknowledgement

Artificial Intelligence (AI) tools were used during the preparation of this report to assist with language refinement, editing, formatting and the organisation of ideas. All experiences, observations, reflections, technical information and conclusions presented in this report are based on the author's own practical training experience, personal reflections and supporting documentation obtained during the placement. The final content was reviewed, edited and verified by the author to ensure its accuracy and authenticity.